

Objective:

The objective of this assessment is to evaluate your understanding and ability to apply supervised learning techniques to a real-world dataset.

Dataset:

Use the breast cancer dataset available in the sklearn library.

Key components to be fulfilled:**1. Loading and Preprocessing (2 marks)**

- Load the breast cancer dataset from sklearn.
- Preprocess the data to handle any missing values and perform necessary feature scaling.
- Explain the preprocessing steps you performed and justify why they are necessary for this dataset.

2. Classification Algorithm Implementation (5 marks)

- Implement the following five classification algorithms:
 1. **Logistic Regression**
 2. **Decision Tree Classifier**
 3. **Random Forest Classifier**
 4. **Support Vector Machine (SVM)**
 5. **k-Nearest Neighbors (k-NN)**
- For each algorithm, provide a brief description of how it works and why it might be suitable for this dataset.

3. Model Comparison (2 marks)

- Compare the performance of the five classification algorithms.
- Which algorithm performed the best and which one performed the worst?

4. Timely Submission (1 mark)**Submission Guidelines:**

- Provide your code in a Jupyter Notebook format and submit the GitHub link here.
- Ensure your explanations and answers are clear and concise.

Total Score: 10